

LAPORAN PRAKTIKUM 1 - STRUKTUR SISTEM OPERASI



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1. Operating-System Services

Praktik 1.1: Menjelajahi Layanan OS

```
putrafermenda@DESKTOP-TTI: ~$ sudo apt install ubuntu-desktop
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: putrafermenda
New password:
Retype new password:
passwd: password updated successfully
Installation successful!
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

putrafermenda@DESKTOP-TTI:~$ cat /etc/os-release
NAME="Ubuntu"
VERSION="24.04.1 LTS"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 24.04.1 LTS"
VERSION_ID="24.04"
BUILD_ID=ubuntu~24.04~1
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/data-policy/privacy-policy"
DEBIAN_FRONTEND=noninteractive

putrafermenda@DESKTOP-TTI:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 21752 12244 ?        Ss   22:19   0:00 /sbin/init
root         2  0.0  0.0   3672 1664 ?        Sl   22:19   0:00 /init
root         7  0.0  0.0   3688 1920 ?        Sl   22:19   0:00 /usr/sbin/plymouthd --control-pocket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        53  0.1  0.2 42160 15624 ?        Ss   22:19   0:00 /usr/lib/systemd/systemd-journald
systemd+  185  0.1  0.0 25268 6272 ?        Ss   22:19   0:00 /usr/lib/systemd/systemd-udev
systemd+  111  0.0  0.1 21852 12072 ?        Ss   22:19   0:00 /usr/lib/systemd/systemd-resolved
systemd+  112  0.0  0.1 91828 7424 ?        Ssl  22:19   0:00 /usr/lib/systemd/systemd-timesyncd
root       197  0.0  0.0 4228 2568 ?        Ss   22:19   0:00 /usr/sbin/cron -f
message+  170  0.0  0.0 9536 4992 ?        Ss   22:19   0:00 /usr/libexec/ssh-keyex --system --address=systemd: --nofork --nopidfile --system-activation --syslog-only
root       186  0.0  0.1 17968 8448 ?        Ss   22:19   0:00 /usr/lib/systemd/systemd-logind
root       197  0.0  0.0 175280 12160 ?       Ssl  22:19   0:00 /usr/libexec/ssh-keyex --system --address=systemd: --nofork --nopidfile --system-activation --syslog-only
root       205  0.0  0.0 3160 1920 hvcb ?    Ss+  22:19   0:00 /usr/sbin/agetty --p --vu --noclear --keep-baud - 115200,38400,9600 vt220
syslog    289  0.0  0.0 222588 5120 ?       Ssl  22:19   0:00 /usr/sbin/rsyslogd -n -iNONE
root       212  0.0  0.0 3116 1792 ttyL    Ss+  22:19   0:00 /usr/sbin/agetty --p --vu --noclear - Linux
root       216  0.0  0.3 186996 22704 ?       Ssl  22:19   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root       271  0.0  0.0   3676 1824 ?        Ss   22:19   0:00 /init
root       272  0.0  0.0   3692 1824 ?        Ss   22:19   0:00 /init
putrafer+ 279  0.0  0.0 6072 4992 pts/0    Ss   22:19   0:00 -bash
root       285  0.0  0.0 6064 4688 pts/1    Ss   22:19   0:00 /bin/login -f

putrafermenda@DESKTOP-TTI:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           6.0Gi        507Mi        5.9Gi         3.6Mi        407Mi        6.1Gi
Swap:          2.0Gi         80k        2.0Gi

putrafermenda@DESKTOP-TTI:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
```

Praktik 1.2: Program C - Menggunakan Layanan OS

Buat file os_services.c:

```
putrafermenda@DESKTOP-TTI: ~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0 388.4M  1 disk
sdb         8:16   0  160M  1 disk
sdc         8:32   0   2G  0 disk [swapon]
sdd         8:48   0   1T  0 disk /mnt/wslg/distro

putrafermenda@DESKTOP-TTI:~$ nano os_services.c
putrafermenda@DESKTOP-TTI:~$ gcc -o os_services os_services.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
putrafermenda@DESKTOP-TTI:~$ sudo apt install gcc
[sudo] password for putrafermenda:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cpp-13 cpp-13-x86-64-linux-gnu gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
  gcc-x86-64-linux-gnu libasan8 libatomic1 libb265-0 libgcc-13-dev libgdb3 libgomp1
  libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1 libhwasan0 libisl23 libitm1
  liblsan0 libmpc3 libquadmath0 libtsan2 libubsan1 libxpm4
  linux-libc-dev manpages-dev rpcsvc-proto
Suggested packages:
  cpp-doc gcc-13-locales cpp-13-doc gcc-multilib make autoconf automake libtool flex bison gnu-objc gcc-13-multilib
  gcc-13-doc gdb-x86-64-linux-gnu glibc-doc libgd-tools libheif-plugin-x265 libheif-plugin-ffmpegdec libheif-plugin-jpegdec libheif-plugin-jpegenc libheif-plugin-j2kdec libheif-plugin-j2kenc libheif-plugin-rav1e libheif-plugin-svtenc
The following NEW packages will be installed:
  cpp-13 cpp-13-x86-64-linux-gnu gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libasan8 libatomic1
  libb265-0 libgcc-13-dev libgdb3 libgomp1 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1 libhwasan0 libisl23 libitm1
  liblsan0 libmpc3 libquadmath0 libtsan2 libubsan1 libxpm4
  linux-libc-dev manpages-dev rpcsvc-proto
0 upgraded, 37 newly installed, 0 to remove and 0 not upgraded.
Need to get 15.0 MB of archives.
After this operation, 180 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 gcc-13-base amd64 13.3.0-6ubuntu2-24.04 [11.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libisl23 amd64 0.26-3build1 [688 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libmpc3 amd64 1.3.1-1build1 [54.5 kB]
```

Compile dan jalankan:

```
putrafernanda@DESKTOP-TTH X + v
Setting up libquadmath0:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libmpc3:amd64 (1.3.1-1build1) ...
Setting up libatomic1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libubsan1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libblas3:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libcrypt-dev:amd64 (1:4.4.36-1build1) ...
Setting up libasan8:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libtsan2:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libisl23:amd64 (0.26-1build1) ...
Setting up libde265-0:amd64 (1.0.15-1build3) ...
Setting up libc-dev-bin (2.39-ubuntu8.6) ...
Setting up libcc1-0:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up liblsan0:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up libitm1:amd64 (14.2.0-ubuntu2-24.04) ...
Setting up cpp-13-x86-64-linux-gnu (13.3.0-ubuntu2-24.04) ...
Setting up libgcc-13-dev:amd64 (13.3.0-ubuntu2-24.04) ...
Setting up libcc-dev:amd64 (2.39-ubuntu8.6) ...
Setting up cpp-13 (13.3.0-ubuntu2-24.04) ...
Setting up gcc-13-x86-64-linux-gnu (13.3.0-ubuntu2-24.04) ...
Setting up gcc-13 (13.3.0-ubuntu2-24.04) ...
Setting up gcc (4:13.2.0-7ubuntu1) ...
Setting up gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up libhif-plugin-asmdec:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libhif:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libg3:amd64 (2.2.3-9ubuntu2) ...
Setting up libc-devtools (2.39-ubuntu8.6) ...
Setting up libhif-plugin-libde265:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libhif-plugin-asmenc:amd64 (1.17.6-1ubuntu4.1) ...
Processing triggers for libc-bin (2.39-ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
putrafernanda@DESKTOP-TTH:~$ gcc -o os_services os_services.c
putrafernanda@DESKTOP-TTH:~$ ./os_services
=== Program Execution Service ===
Process ID (PID): 8156
Parent Process ID (PPID): 279

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: g+w

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
```

2. User and Operating-System Interface

Praktik 2.1: Command Line Interface (CLI)

```
putrafernanda@DESKTOP-TTH X + v
Parent Process ID (PPID): 279

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: g+w

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
putrafernanda@DESKTOP-TTH:~$ pwd
/home/putrafernanda
putrafernanda@DESKTOP-TTH:~$ ls -la
total 56
drwxr-xr-x 4 putrafernanda putrafernanda 4096 Oct 13 23:03 .
drwxr-xr-x 3 root root 4096 Oct 13 22:19 ..
-rw-r--r-- 1 putrafernanda putrafernanda 228 Oct 13 22:19 .bash_logout
-rw-r--r-- 1 putrafernanda putrafernanda 3771 Oct 13 22:19 .bashrc
drwx----- 2 putrafernanda putrafernanda 4096 Oct 13 22:19 .cache
drwxr-xr-x 3 putrafernanda putrafernanda 4096 Oct 13 22:25 .local
-rw-rw-r-- 1 putrafernanda putrafernanda 0 Oct 13 22:19 .netdShown
-rw-r--r-- 1 putrafernanda putrafernanda 887 Oct 13 22:19 .profile
-rw-r--r-- 1 putrafernanda putrafernanda 0 Oct 13 22:28 .sudo_as_admin_successful
-rwxr-xr-x 1 putrafernanda putrafernanda 16488 Oct 13 23:02 os_services
-rw-r--r-- 1 putrafernanda putrafernanda 1315 Oct 13 22:26 os_services.c
-rw-r--r-- 1 putrafernanda putrafernanda 23 Oct 13 23:03 test_file.txt
putrafernanda@DESKTOP-TTH:~$ cd /tmp
putrafernanda@DESKTOP-TTH:~/tmp$ mkdir test_dir
putrafernanda@DESKTOP-TTH:~/tmp$ cd test_dir
putrafernanda@DESKTOP-TTH:~/tmp/test_dir$ cd
putrafernanda@DESKTOP-TTH:~$ ls -l | grep "d"
putrafernanda@DESKTOP-TTH:~$ ps aux | grep bash | wc -l
3
putrafernanda@DESKTOP-TTH:~$ echo "Hello World" > output.txt
cat output.txt
cat < output.txt
cat << output.txt
Hello World
cat output.txt: input file is output file
Hello World
putrafernanda@DESKTOP-TTH:~$ sleep 30 &
[1] 8173
putrafernanda@DESKTOP-TTH:~$ jobs
[1]+  Running                  sleep 30 &
putrafernanda@DESKTOP-TTH:~$ fg %1
sleep 30
```

Praktik 2.2: Program C - Shell Sederhana

Buat file `simple_shell.c`:

```
putrafernanda@DESKTOP-TTKU011:~$ nano simple_shell.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o simple_shell simple_shell.c
putrafernanda@DESKTOP-TTKU011:~$ ./simple_shell
myshell> ^C
putrafernanda@DESKTOP-TTKU011:~$ nano process_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ nano file_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ nano info_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o process_syscalls process_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o file_syscalls file_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o info_syscalls info_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 8244
Parent waiting for child (PID: 8245)...
Child PID: 8245
Child's Parent PID: 8244
Child executing 'ls' command...
total 116
-rwxr-xr-x 1 putrafernanda putrafernanda 16352 Oct 13 23:44 file_syscalls
-rw-r--r-- 1 putrafernanda putrafernanda 966 Oct 13 23:41 file_syscalls.c
-rwxr-xr-x 1 putrafernanda putrafernanda 16384 Oct 13 23:44 info_syscalls
-rw-r--r-- 1 putrafernanda putrafernanda 711 Oct 13 23:43 info_syscalls.c
-rwxr-xr-x 1 putrafernanda putrafernanda 16488 Oct 13 23:02 os_services
-rw-r--r-- 1 putrafernanda putrafernanda 1315 Oct 13 22:26 os_services.c
-rw-r--r-- 1 putrafernanda putrafernanda 12 Oct 13 23:06 output.txt
-rwxr-xr-x 1 putrafernanda putrafernanda 16360 Oct 13 23:43 process_syscalls
-rw-r--r-- 1 putrafernanda putrafernanda 831 Oct 13 23:40 process_syscalls.c
-rwxr-xr-x 1 putrafernanda putrafernanda 16520 Oct 13 23:08 simple_shell
-rw-r--r-- 1 putrafernanda putrafernanda 897 Oct 13 23:08 simple_shell.c
-rw-r--r-- 1 putrafernanda putrafernanda 23 Oct 13 23:03 test_file.txt
Child exited with status: 0
```

3. System Calls Praktik

3.1: Process Control System Calls

Buat file `process_syscalls.c`:

Praktik 3.2: File Management System Calls

Buat file `file_syscalls.c`:

Praktik 3.3: Information Maintenance System Calls

Buat file `info_syscalls.c`:

Compile semua:

```
putrafernanda@DESKTOP-TTKU011:~$ jobs
[1]+  Running                  sleep 30 &
putrafernanda@DESKTOP-TTKU011:~$ fg %1
sleep 30
putrafernanda@DESKTOP-TTKU011:~$ nano simple_shell.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o simple_shell simple_shell.c
putrafernanda@DESKTOP-TTKU011:~$ ./simple_shell
myshell> ^C
putrafernanda@DESKTOP-TTKU011:~$ nano process_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ nano file_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ nano info_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o process_syscalls process_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o file_syscalls file_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o info_syscalls info_syscalls.c
putrafernanda@DESKTOP-TTKU011:~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 8244
Parent waiting for child (PID: 8245)...
Child PID: 8245
Child's Parent PID: 8244
Child executing 'ls' command...
total 116
-rwxr-xr-x 1 putrafernanda putrafernanda 16352 Oct 13 23:44 file_syscalls
-rw-r--r-- 1 putrafernanda putrafernanda 966 Oct 13 23:41 file_syscalls.c
-rwxr-xr-x 1 putrafernanda putrafernanda 16384 Oct 13 23:44 info_syscalls
-rw-r--r-- 1 putrafernanda putrafernanda 711 Oct 13 23:43 info_syscalls.c
-rwxr-xr-x 1 putrafernanda putrafernanda 16488 Oct 13 23:02 os_services
-rw-r--r-- 1 putrafernanda putrafernanda 1315 Oct 13 22:26 os_services.c
-rw-r--r-- 1 putrafernanda putrafernanda 12 Oct 13 23:06 output.txt
-rwxr-xr-x 1 putrafernanda putrafernanda 16360 Oct 13 23:43 process_syscalls
-rw-r--r-- 1 putrafernanda putrafernanda 831 Oct 13 23:40 process_syscalls.c
-rwxr-xr-x 1 putrafernanda putrafernanda 16520 Oct 13 23:08 simple_shell
-rw-r--r-- 1 putrafernanda putrafernanda 897 Oct 13 23:08 simple_shell.c
-rw-r--r-- 1 putrafernanda putrafernanda 23 Oct 13 23:03 test_file.txt
Child exited with status: 0
putrafernanda@DESKTOP-TTKU011:~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
putrafernanda@DESKTOP-TTKU011:~$ ./info_syscalls
=== Information Maintenance System Calls ===
Process ID: 8247
Parent Process ID: 279
User ID: 1000
Group ID: 1000
Current time: Mon Oct 13 23:44:40 2025
Clock ticks since boot: 1718457561
Sleeping for 2 seconds...
Awake!
putrafernanda@DESKTOP-TTKU011:~$ |
```

4. System Services

Praktik 4.1: Menggunakan systemd

```
putrafernanda@DESKTOP-TTKU011:~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-getty.service              loaded active running Console Getty
console-setup.service              loaded active exited Set console font and keymap
cron.service                        loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service             loaded active exited Set the console keyboard layout
kmod-static-nodes.service          loaded active exited Create List of Static Device Nodes
my-daemon.service                  loaded active running My Custom Daemon
polkit.service                     loaded active running Authorization Manager
rsyslog.service                    loaded active running System Logging Service
setvtrgb.service                   loaded active exited Set console scheme
snapd.seeded.service              loaded active exited Wait until snapd is fully seeded
systemd-binfmt.service             loaded active exited Set Up Additional Binary Formats
systemd-journal-flush.service      loaded active exited Flush Journal to Persistent Storage
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-modules-load.service        loaded active exited Load Kernel Modules
systemd-remount-fs.service          loaded active exited Remount Root and Kernel File Systems
systemd-resolved.service           loaded active running Network Name Resolution
systemd-sysctl.service             loaded active exited Apply Kernel Variables
systemd-timesyncd.service           loaded active running Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service loaded active exited Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service     loaded active exited Create Volatile Files and Directories
systemd-udev-trigger.service        loaded active exited Coldplug All udev Devices
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
systemd-update-utmp.service         loaded active exited Record System Boot/Shutdown in UTMP
systemd-user-sessions.service       loaded active exited Permit User Sessions

putrafernanda@DESKTOP-TTKU011:~$ systemctl status ssh
Unit ssh.service could not be found.
putrafernanda@DESKTOP-TTKU011:~$ sudo apt update
[sudo] password for putrafernanda:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
```

```
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
putrafernanda@DESKTOP-TTKU011:~$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; vendor preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)

putrafernanda@DESKTOP-TTKU011:~$ systemctl status ssh.socket
● ssh.socket - OpenBSD Secure Shell server socket
   Loaded: loaded (/usr/lib/systemd/system/ssh.socket; enabled; vendor preset: enabled)
   Active: active (listening) since Tue 2025-10-14 18:13:50 WIB; 1ms ago
   Triggers: ● ssh.service
   Listen: 0.0.0.0:22 (Stream)
           [::]:22 (Stream)
   Tasks: 0 (limit: 8152)
   Memory: 12.0K (peak: 264.0K)
   CPU: 2ms
   CGroup: /system.slice/ssh.socket

Oct 14 18:13:50 DESKTOP-TTKU011 systemd[1]: Listening on ssh.socket.
putrafernanda@DESKTOP-TTKU011:~$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; vendor preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)

putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl disable ssh.socket
Removed "/etc/systemd/system/ssh.service.requires/ssh.socket".
Removed "/etc/systemd/system/sockets.target.wants/ssh.socket".
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl start ssh
putrafernanda@DESKTOP-TTKU011:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
```

```
putrafernanda@DESKTOP-TTKU011:~$ journalctl -u ssh -n 20
Oct 14 18:17:01 DESKTOP-TTKU011 systemd[1]: Starting ssh.service - Open
Oct 14 18:17:01 DESKTOP-TTKU011 sshd[2568]: Server listening on 0.0.0.0
Oct 14 18:17:01 DESKTOP-TTKU011 sshd[2568]: Server listening on :: port
Oct 14 18:17:01 DESKTOP-TTKU011 systemd[1]: Started ssh.service - Open
putrafernanda@DESKTOP-TTKU011:~$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE    PRESET
apparmor.service                        enabled enabled
apport.service                          enabled enabled
cloud-config.service                    enabled enabled
cloud-final.service                      enabled enabled
cloud-init-local.service                  enabled enabled
cloud-init.service                       enabled enabled
console-setup.service                    enabled enabled
cron.service                            enabled enabled
dmesg.service                            enabled enabled
e2scrub_reap.service                     enabled enabled
getty@.service                           enabled enabled
keyboard-setup.service                   enabled enabled
landscape-client.service                  enabled enabled
my_daemon.service                        enabled enabled
networkd-dispatcher.service              enabled enabled
rsyslog.service                          enabled enabled
setvtrgb.service                         enabled enabled
snapd.apparmor.service                   enabled enabled
snapd.autoimport.service                  enabled enabled
snapd.core-fixup.service                  enabled enabled
snapd.recovery-chooser-trigger.service    enabled enabled
snapd.seeded.service                     enabled enabled
snapd.service                            enabled enabled
snapd.system-shutdown.service             enabled enabled
ssh.service                              enabled enabled
systemd-pstore.service                    enabled enabled
systemd-resolved.service                  enabled enabled
systemd-timesyncd.service                  enabled enabled
ua-reboot-cmds.service                    enabled enabled
ubuntu-advantage.service                  enabled enabled
unattended-upgrades.service               enabled enabled
wsl-pro.service                           enabled enabled
```

Praktik 4.2: Membuat Custom System Service

Buat program sederhana `my_daemon.c`:

Compile:

Buat service file `/etc/systemd/system/my_daemon.service`:

Kelola service:

```
putrafernanda@DESKTOP-TTKU011:~$ nano my_daemon.c
putrafernanda@DESKTOP-TTKU011:~$ gcc -o my_daemon my_daemon.c
putrafernanda@DESKTOP-TTKU011:~$ sudo nano /etc/systemd/system/my_daemon.service
[sudo] password for putrafernanda:
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl daemon-reload
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl start my_daemon
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl status my_daemon
● my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-10-14 18:09:08 WIB; 1min 1s ago
     Main PID: 1743 (python3)
       Tasks: 1 (Limit: 8152)
      Memory: 5.1M (peak: 5.4M)
         CPU: 94ms
    CGroup: /system.slice/my_daemon.service
            └─1743 /usr/bin/python3 /home/putrafernanda/my_pro

Oct 14 18:09:08 DESKTOP-TTKU011 systemd[1]: Started my_daemon.service.
putrafernanda@DESKTOP-TTKU011:~$ sudo journalctl -u my_daemon
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: /etc/systemd/system/my_daemon.service:1: Unit my_daemon.service is not loaded.
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: /etc/systemd/system/my_daemon.service:1: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: Starting my_daemon.service.
Oct 14 17:09:13 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:13 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:14 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:14 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:14 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:14 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:14 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:14 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:15 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:15 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
Oct 14 17:09:15 DESKTOP-TTKU011 systemd[1]: my_daemon.service: Failed to start my_daemon.service: Unit my_daemon.service is not loaded.
```

```
putrafernanda@DESKTOP-TTKU011:~$ ls -l /home/putrafernanda/my_daemon
-rwxr-xr-x 1 putrafernanda putrafernanda 16344 Oct 14 18:36 /home/putrafernanda/my_daemon
putrafernanda@DESKTOP-TTKU011:~$ ls -l /home/putrafernanda/my_daemon
-rwxr-xr-x 1 putrafernanda putrafernanda 16344 Oct 14 18:36 /home/putrafernanda/my_daemon
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl daemon-reload
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl enable my_daemon.service
putrafernanda@DESKTOP-TTKU011:~$ sudo systemctl start my_daemon.service
putrafernanda@DESKTOP-TTKU011:~$ systemctl status my_daemon.service
● my_daemon.service My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-10-14 18:09:08 WIB; 33min ago
     Main PID: 1743 (python3)
       Tasks: 1 (Limit: 8152)
      Memory: 5.1M (peak: 5.4M)
         CPU: 103ms
        CGroup: /system.slice/my_daemon.service
                └─1743 /usr/bin/python3 /home/putrafernanda/my_project/my_daemon.py

Oct 14 18:09:08 DESKTOP-TTKU011 systemd[1]: Started my_daemon.service - My Custom Daemon.
putrafernanda@DESKTOP-TTKU011:~$ sudo journalctl -u my_daemon
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: /etc/systemd/system/my_daemon.service:1: Assignment outside of section. Ignoring.
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: /etc/systemd/system/my_daemon.service:2: Assignment outside of section. Ignoring.
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: /etc/systemd/system/my_daemon.service:3: Assignment outside of section. Ignoring.
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: /etc/systemd/system/my_daemon.service:4: Assignment outside of section. Ignoring.
Oct 14 17:00:22 DESKTOP-TTKU011 systemd[1]: /etc/systemd/system/my_daemon.service:5: Assignment outside of section. Ignoring.
```

5. Linkers and Loaders

Praktik 5.1: Proses Linking

Buat tiga file:

math_ops.h:

math_ops.c:

main.c:

Proses kompilasi bertahap:

```
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ nano math_ops.h
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ nano math_ops.c
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ nano main.c
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ gcc -E math_ops.c -o math_ops.i
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ gcc -E main.c -o main.i
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ gcc -E math_ops.c -o math_ops.s
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ gcc -S main.c -o main.s
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ gcc -c math_ops.c -o math_ops.o
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ gcc -c main.c -o main.o
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ gcc main.o math_ops.o -o program
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ nm main.o
0000000000000000 T main
0000000000000000 U multiply
0000000000000000 U printf
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ nm math_ops.o
0000000000000000 T add
0000000000000010 T multiply
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
0000000000002148 w _ITM_deregisterTMCloneTable
0000000000002148 w _ITM_registerTMCloneTable
0000000000002148 r _FRAME_END_
000000000000201c r _GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
00000000000038c r __abi_tag
0000000000004010 B __bss_start
0000000000004000 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004008 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
0000000000004000 w __gmon_start__
0000000000004010 U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
```



```
putrafernanda@DESKTOP-TTI X + v
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
w _ITM_deregisterTMCloneTable
w _ITM_registerTMCloneTable
0000000000002148 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000033c r __abi_tag
0000000000004010 B __bss_start
w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_ctors_aux
0000000000003dc0 d __do_global_ctors_aux_fini_array_entry
0000000000004008 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
w __gmon_start__
U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ ldd program
linux-vdso.so.1 (0x00007ffd35176000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000077eed8600000)
/lib64/ld-linux-x86-64.so.2 (0x000077eed88c1000)
putrafernanda@DESKTOP-TTKU011:~/linker_praktik$ cd
putrafernanda@DESKTOP-TTKU011:~$
```

Praktik 5.2: Static vs Dynamic Linking

Praktik 5.3: Membuat dan Menggunakan Shared Library

```
putrafernanda@DESKTOP-TTI X + v
putrafernanda@DESKTOP-TTKU011:~$ gcc main.c math_ops.c -o program_static -static
putrafernanda@DESKTOP-TTKU011:~$ gcc main.c math_ops.c -o program_dynamic
putrafernanda@DESKTOP-TTKU011:~$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 putrafernanda putrafernanda 16K Oct 14 19:24 program_dynamic
-rwxr-xr-x 1 putrafernanda putrafernanda 767K Oct 14 19:24 program_static
putrafernanda@DESKTOP-TTKU011:~$ ldd program_static
not a dynamic executable
putrafernanda@DESKTOP-TTKU011:~$ ldd program_dynamic
not a dynamic executable
putrafernanda@DESKTOP-TTKU011:~$ gcc -c -fPIC math_ops.c -o math_ops.o
putrafernanda@DESKTOP-TTKU011:~$ gcc -shared -o libmathops.so math_ops.o
putrafernanda@DESKTOP-TTKU011:~$ gcc main.c -L. -lmathops -o program_shared
putrafernanda@DESKTOP-TTKU011:~$ LD_LIBRARY_PATH=. ./program_shared
3 + 2 = 5
3 - 2 = 1
putrafernanda@DESKTOP-TTKU011:~$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
Tag                Type                Name/Value
0x0000000000000001 (NEEDED)            Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)            Shared library: [libc.so.6]
0x000000000000000c (INIT)              0x1000
0x000000000000000d (FINI)              0x11e4
0x0000000000000010 (INIT_ARRAY)        0x3d98
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)        0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x000000000000ffff5 (GNU_HASH)          0x3b0
0x0000000000000005 (STRTAB)             0x4b0
0x0000000000000006 (SYMTAB)             0x3d8
0x000000000000000a (STRSZ)              170 (bytes)
0x000000000000000b (SYMINFO)            24 (bytes)
0x0000000000000015 (DEBUG)              0x0
0x0000000000000003 (PLTGOT)            0x3fa8
0x0000000000000002 (PLTRELSZ)          72 (bytes)
0x0000000000000014 (PLTREL)             RELA
0x0000000000000017 (JMPREL)            0x660
0x0000000000000007 (RELA)              0x5a0
0x0000000000000008 (RELASZ)            192 (bytes)
0x0000000000000009 (RELAENT)           24 (bytes)
0x000000000000001e (FLAGS)              BIND_NOW
0x000000000000ffffb (FLAGS_1)          Flags: NOW PIE
0x000000000000ffffe (VERNEED)          0x570
```

6. Why Applications Are Operating-System Specific

Praktik 6.1: System Call Differences

Buat file `os_specific.c`:

Praktik 6.2: Binary Format Analysis

```
putrafernanda@DESKTOP-TTHU011:~$ nano os_specific.c
putrafernanda@DESKTOP-TTHU011:~$ ^C
putrafernanda@DESKTOP-TTHU011:~$ gcc os_specific.c -o program
putrafernanda@DESKTOP-TTHU011:~$ ls
file_syscalls  math_ops.h      process_syscalls.c
file_syscalls.c math_ops.o      program
info_syscalls  my_daemon       program_dynamic
info_syscalls.c my_daemon.c     program_shared
libmathops.so  my_project      program_static
linker_praktik os_services     simple_shell
main.c         os_services.c   simple_shell.c
main.i         os_specific.c   syscall_test.txt
math_ops.c     output.txt      test_file.txt
math_ops.c: process_syscalls
putrafernanda@DESKTOP-TTHU011:~$ ./program
=== OS-Specific Application ===
OS: Linux
System: Linux
Release: 6.6.87.2-microsoft-standard-WSL2
Version: #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025

Linux-specific: Process ID = 448
putrafernanda@DESKTOP-TTHU011:~$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=334d4d381bcb2c27c8d33b5f83d6d9831f888181, for GNU/Linux 3.2.0, not stripped
putrafernanda@DESKTOP-TTHU011:~$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 02 01 01 00 00 00 00 00 00 00 00 00 00
  Class:             ELF64
  Data:              2's complement, little endian
  Version:           1 (current)
  OS/ABI:            UNIX - System V
  ABI Version:       0
  Type:              DYN (Position-Independent Executable file)
  Machine:           Advanced Micro Devices X86-64
  Version:           0x1
  Entry point address: 0x10e0
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14208 (bytes into file)
  Flags:             0x0
```

```
putrafernanda@DESKTOP-TTHU011:~$ readelf -S program
Number of section headers: 31
Section header string table index: 30
putrafernanda@DESKTOP-TTHU011:~$ readelf -S program
There are 31 section headers, starting at offset 0x3778:

Section Headers:
[Nr] Name              Type              Address            Offset
     Size              EntSize          Flags Link Info  Align
[ 0] 0000000000000000 NULL              0000000000000000 00000000
[ 1] .interp              PROGBITS          0000000000000318 00000318
     000000000000001c 0000000000000000 A 0 0 1
[ 2] .note.gnu.pr[...]    NOTE              0000000000000338 00000338
     0000000000000030 0000000000000000 A 0 0 8
[ 3] .note.gnu.bu[...]    NOTE              0000000000000368 00000368
     0000000000000024 0000000000000000 A 0 0 4
[ 4] .note.ABI-tag        NOTE              000000000000038c 0000038c
     0000000000000020 0000000000000000 A 0 0 4
[ 5] .gnu.hash             GNU_HASH          00000000000003b0 000003b0
     0000000000000024 0000000000000000 A 6 0 8
[ 6] .dynsym               DYNSYM            00000000000003d8 000003d8
     0000000000000108 0000000000000018 A 7 1 8
[ 7] .dynstr               STRTAB            00000000000004e0 000004e0
     00000000000000bc 0000000000000000 A 0 0 1
[ 8] .gnu.version          VERSYM            000000000000059c 0000059c
     0000000000000016 0000000000000002 A 6 0 2
[ 9] .gnu.version_r        VERNEED           00000000000005b8 000005b8
     0000000000000040 0000000000000000 A 7 1 8
[10] .rela.dyn             RELA              00000000000005f8 000005f8
     00000000000000c0 0000000000000018 A 6 0 8
[11] .rela.plt             RELA              00000000000006b8 000006b8
     0000000000000078 0000000000000018 AI 6 24 8
[12] .init                 PROGBITS          0000000000001000 00001000
     000000000000001b 0000000000000000 AX 0 0 4
[13] .plt                  PROGBITS          0000000000001020 00001020
     0000000000000060 0000000000000010 AX 0 0 16
[14] .plt.got              PROGBITS          0000000000001080 00001080
     0000000000000010 0000000000000010 AX 0 0 16
[15] .plt.sec              PROGBITS          0000000000001090 00001090
     0000000000000050 0000000000000010 AX 0 0 16
```

```

putrafernanda@DESKTOP-TT1: ~
Number of section headers: 31
Section header string table index: 30
putrafernanda@DESKTOP-TT1:~$ readelf -S program
There are 31 section headers, starting at offset 0x3778:

Section Headers:
[Nr] Name           Type              Address            Offset
    Size              EntSize          Flags Link Info Align
[ 0]                  NULL              0000000000000000  0 0 0
0000000000000000  0000000000000000
[ 1] .interp            PROGBITS          0000000000000318  0 0 1
000000000000001c  0000000000000000
[ 2] .note.gnu.pr[...] NOTE              0000000000000338  0 0 8
0000000000000030  0000000000000000
[ 3] .note.gnu.bu[...] NOTE              0000000000000368  0 0 4
0000000000000024  0000000000000000
[ 4] .note.ABI-tag      NOTE              000000000000038c  0 0 4
0000000000000020  0000000000000000
[ 5] .gnu.hash           GNU_HASH          00000000000003b0  0 0 8
0000000000000024  0000000000000000
[ 6] .dynsym             DYNSYM            00000000000003d8  0 0 8
0000000000000108  0000000000000018
[ 7] .dynstr             STRTAB            00000000000004e0  0 0 1
00000000000000bc  0000000000000000
[ 8] .gnu.version        VERSYM            000000000000059c  0 0 2
0000000000000016  0000000000000002
[ 9] .gnu.version_r      VERNEED           00000000000005b8  0 0 8
0000000000000040  0000000000000000
[10] .rela.dyn            RELA              00000000000005f8  0 0 8
00000000000000c0  0000000000000018
[11] .rela.plt            RELA              00000000000006b8  0 0 8
0000000000000078  0000000000000018
[12] .init               PROGBITS          0000000000001000  0 0 4
000000000000001b  0000000000000000
[13] .plt                PROGBITS          0000000000001020  0 0 16
0000000000000060  0000000000000010
[14] .plt.got            PROGBITS          0000000000001080  0 0 16
0000000000000010  0000000000000010
[15] .plt.sec            PROGBITS          0000000000001090  0 0 16
0000000000000050  0000000000000010

```

```

putrafernanda@DESKTOP-TT1: ~
11
12 .init_array .fini_array .dynamic .got
putrafernanda@DESKTOP-TT1:~$ objdump -d program | head -50
program:      file format elf64-x86-64

Disassembly of section .init:

0000000000001000 <.init>:
1000: f3 0f 1e fa          endbr64
1004: 48 83 ec 08          sub    $0x8,%rsp
1008: 48 8b 05 d9 2f 00 00 mov    0x2fd9(%rip),%rax        # 3fe8 <__gmon_start__@Base>
100f: 48 83 c0             test   %rax,%rax
1012: 74 02              je     1016 <.init+0x16>
1014: ff d0              call   *%rax
1016: 48 83 c4 08          add    $0x8,%rsp
101a: c3                 ret

Disassembly of section .plt:

0000000000001020 <.plt>:
1020: ff 35 7a 2f 00 00    push   0x2f7a(%rip)             # 3fa0 <_GLOBAL_OFFSET_TABLE_+0x8>
1026: ff 25 7c 2f 00 00    jmp    +0x2f7c(%rip)            # 3fa8 <_GLOBAL_OFFSET_TABLE_+0x10>
102c: 0f 1f 40 00          nopl   0x0(%rax)
1030: f3 0f 1e fa          endbr64
1034: 60 00 00 00 00 00    push   $0x0
1039: e9 e2 ff ff ff      jmp     1020 <.init+0x20>
103e: 66 90              xchg   %ax,%ax
1040: f3 0f 1e fa          endbr64
1044: 68 01 00 00 00 00    push   $0x1
1049: e9 02 ff ff ff      jmp     1020 <.init+0x20>
104e: 66 90              xchg   %ax,%ax
1050: f3 0f 1e fa          endbr64
1054: 60 02 00 00 00 00    push   $0x2
1059: e9 c2 ff ff ff      jmp     1020 <.init+0x20>
105e: 66 90              xchg   %ax,%ax
1060: f3 0f 1e fa          endbr64
1064: 68 03 00 00 00 00    push   $0x3
1069: e9 b2 ff ff ff      jmp     1020 <.init+0x20>
106e: 66 90              xchg   %ax,%ax
1070: f3 0f 1e fa          endbr64
1074: 68 04 00 00 00 00    push   $0x4
1079: e9 a2 ff ff ff      jmp     1020 <.init+0x20>
107e: 66 90              xchg   %ax,%ax

Disassembly of section .plt.got:

0000000000001080 <_cxa_finalize@plt>:

```

7. OS Design and Implementation

Praktik 7.1: Kernel Module Sederhana

Buat file `hello_module.c`:

Makefile:

Compile dan load:

Praktik 7.2: Monitoring Kernel

```
putrafernanda@DESKTOP-TTU011:~$ nano hello_module.c
putrafernanda@DESKTOP-TTU011:~$ nano Makefile
putrafernanda@DESKTOP-TTU011:~$ make
sudo insmod hello_module.ko
lsmod | grep hello
dmesg | tail
sudo rmmod hello_module
dmesg | tail
Makefile:3: *** missing separator. Stop.
[sudo] password for putrafernanda:
insmod: ERROR: could not insert module hello_module.ko: Invalid module format
[10246.421663] misc dxg: dxgk: dxgkio_query_adapter_info: ioctl failed: -2
[10246.617800] Failed to connect to bus: No such file or directory
[10246.796103] systemd-journald[54]: Collecting audit messages is disabled.
[10246.877083] systemd-journald[54]: Received client request to flush runtime journal.
[10246.878256] systemd-journald[54]: File /var/log/journal/3ce0ba065fcf4f64b99d883ecddabb88/system.journal corrupted or uncleanly shut down, renaming and replacing.
[10248.964477] systemd-journald[54]: /var/log/journal/3ce0ba065fcf4f64b99d883ecddabb88/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[10294.187421] mini_init (219): drop_caches: 1
[11524.457650] mini_init (219): drop_caches: 1
[12762.127725] Invalid ELF header len 35
[12959.034475] Invalid ELF header len 35
rmmod: ERROR: Module hello_module is not currently loaded
[10246.421663] misc dxg: dxgk: dxgkio_query_adapter_info: ioctl failed: -2
[10246.617800] Failed to connect to bus: No such file or directory
[10246.796103] systemd-journald[54]: Collecting audit messages is disabled.
[10246.877083] systemd-journald[54]: Received client request to flush runtime journal.
[10246.878256] systemd-journald[54]: File /var/log/journal/3ce0ba065fcf4f64b99d883ecddabb88/system.journal corrupted or uncleanly shut down, renaming and replacing.
[10248.964477] systemd-journald[54]: /var/log/journal/3ce0ba065fcf4f64b99d883ecddabb88/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[10294.187421] mini_init (219): drop_caches: 1
[11524.457650] mini_init (219): drop_caches: 1
[12762.127725] Invalid ELF header len 35
[12959.034475] Invalid ELF header len 35
putrafernanda@DESKTOP-TTU011:~$ make
Makefile:3: *** missing separator. Stop.
putrafernanda@DESKTOP-TTU011:~$ nano Makefile
putrafernanda@DESKTOP-TTU011:~$ make
```

```
putrafernanda@DESKTOP-TTU011:~$ uname -r
6.6.87.2-microsoft-standard-WSL2
putrafernanda@DESKTOP-TTU011:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025
putrafernanda@DESKTOP-TTU011:~$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall2 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
putrafernanda@DESKTOP-TTU011:~$ dmesg | tail -30
[ 9575.529053] misc dxg: dxgk: dxgkio_query_adapter_info: ioctl failed: -22
[ 9575.529089] misc dxg: dxgk: dxgkio_query_adapter_info: ioctl failed: -2
[ 9575.715034] Failed to connect to bus: No such file or directory
[ 9575.911048] systemd-journald[54]: Collecting audit messages is disabled.
[ 9575.91374] systemd-journald[54]: Received client request to flush runtime journal.
[ 9575.993949] systemd-journald[54]: File /var/log/journal/3ce0ba065fcf4f64b99d883ecddabb88/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 9578.123766] systemd-journald[54]: /var/log/journal/3ce0ba065fcf4f64b99d883ecddabb88/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 9634.299848] mini_init (219): drop_caches: 1
```

```
putrafernanda@DESKTOP-TTI: ~$ cat /var/log/journal/3ce8a065fcf4f64b99d883ecddabb88/user-1000.journal: Journal file uses a different sequence number 10, rotating.
[10294.964477] systemd-journal[54]: /var/log/journal/3ce8a065fcf4f64b99d883ecddabb88/user-1000.journal: Journal file uses a different sequence number 10, rotating.
[10294.187421] mini_init (219): drop_caches: 1
[11524.457650] mini_init (219): drop_caches: 1
[12762.127725] Invalid ELF header len 35
[12959.034475] Invalid ELF header len 35
[13355.394558] mini_init (219): drop_caches: 1
[13481.292228] Invalid ELF header len 35
[13503.736373] Invalid ELF header len 35
putrafernanda@DESKTOP-TTU011:~$ lsmod
Module                  Size  Used by
tls                     98384  0
intel_rapl_msr          16384  0
intel_rapl_common       32768  1 intel_rapl_msr
kvm_amd                 126976  0
kvm                     970752  1 kvm_amd
irqbypass              12288  1 kvm
crc32c_intel            16384  2
battery                 20480  0
ac                      16384  0
sch_fq_codel            16384  1
configfs                53248  1
autofs4                 45056  0
br_netfilter            28672  0
bridge                 282624  1 br_netfilter
stp                     12288  1 bridge
llc                     12288  2 bridge, stp
ip_tables               28672  0
tun                     53248  0
putrafernanda@DESKTOP-TTU011:~$ modinfo ext4
name:                   ext4
filename:               (builtin)
softdep:                pre: crc32c
license:                GPL
file:                   fs/ext4/ext4
description:            Fourth Extended Filesystem
author:                 Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias:                  fs-ext4
alias:                  ext3
alias:                  fs-ext3
```

8. Operating-System Structure

Praktik 8.1: Eksplorasi Struktur Layering

Praktik 8.2: Microkernel vs Monolithic

Praktik 8.3: Program - Virtual File System

Compile dan jalankan:

```
putrafernanda@DESKTOP-TTI: ~$ alias: fs-ext3
putrafernanda@DESKTOP-TTU011:~$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
putrafernanda@DESKTOP-TTU011:~$ sudo apt install pciutils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-glx
  libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libpci3
Suggested packages:
  bzip2
The following NEW packages will be installed:
  libpci3 pciutils
0 upgraded, 2 newly installed, 0 to remove and 2 not upgraded.
Need to get 186 kB of archives.
After this operation, 301 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libpci3 amd64 1:3.10.0-2build1 [36.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 pciutils amd64 1:3.10.0-2build1 [69.7 kB]
Fetched 186 kB in 2s (45.3 kB/s)
Selecting previously unselected package libpci3:amd64.
(Reading database ... 48857 files and directories currently installed.)
Preparing to unpack .../libpci3_1:3.10.0-2build1_amd64.deb ...
Unpacking libpci3:amd64 (1:3.10.0-2build1) ...
Selecting previously unselected package pciutils.
Preparing to unpack .../pciutils_1:3.10.0-2build1_amd64.deb ...
Unpacking pciutils (1:3.10.0-2build1) ...
Setting up libpci3:amd64 (1:3.10.0-2build1) ...
Setting up pciutils (1:3.10.0-2build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.0) ...
putrafernanda@DESKTOP-TTU011:~$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
putrafernanda@DESKTOP-TTU011:~$ sudo apt install usbutils
```

```
putrafernanda@DESKTOP-TT1: ~$ sudo apt install usbutils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-gli
  libglapi-mesa libllvm1764 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libusb-1.0-0
Suggested packages:
  usbutils-py
The following NEW packages will be installed:
  libusb-1.0-0 usbutils
0 upgraded, 2 newly installed, 0 to remove and 2 not upgraded.
Need to get 159 kB of archives.
After this operation, 469 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libusb-1.0-0 amd64 2:1.0.27-1 [54.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 usbutils amd64 1:017-3build1 [84.5 kB]
Fetched 159 kB in 2s (81.3 kB/s)
Selecting previously unselected package libusb-1.0-0:amd64.
(Reading database ... 48075 files and directories currently installed.)
Preparing to unpack .../libusb-1.0-0_2:1.0.27-1_amd64.deb ...
Unpacking libusb-1.0-0:amd64 (2:1.0.27-1) ...
Selecting previously unselected package usbutils.
Preparing to unpack .../usbutils_1:017-3build1_amd64.deb ...
Unpacking usbutils (1:017-3build1) ...
Setting up libusb-1.0-0:amd64 (2:1.0.27-1) ...
Setting up usbutils (1:017-3build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
putrafernanda@DESKTOP-TT1: ~$ cat /proc/cpuinfo
processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 25
model          : 68
model name     : AMD Ryzen 5 6600H with Radeon Graphics
stepping       : 1
microcode     : 0xffffffff
cpu MHz        : 3293.811
cache size     : 512 KB
physical id    : 0
siblings       : 12
core id        : 0
cpu cores      : 6
apicid         : 0
initial apicid : 0
```

```
cache_alignment : 64
address sizes    : 48 bits physical, 48 bits virtual
power management:

putrafernanda@DESKTOP-TT1: ~$ uname -a
Linux DESKTOP-TT1 6.8.0-26-generic-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 GNU/Linux
putrafernanda@DESKTOP-TT1: ~$ cat /proc/sys/kernel/ostype
Linux
putrafernanda@DESKTOP-TT1: ~$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
putrafernanda@DESKTOP-TT1: ~$ sudo apt install strace
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-gli
  libglapi-mesa libllvm1764 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  strace
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 584 kB of archives.
After this operation, 2153 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 strace amd64 6.8-0ubuntu2 [584 kB]
Fetched 584 kB in 2s (289 kB/s)
Selecting previously unselected package strace.
(Reading database ... 48094 files and directories currently installed.)
Preparing to unpack .../strace_6.8-0ubuntu2_amd64.deb ...
Unpacking strace (6.8-0ubuntu2) ...
Setting up strace (6.8-0ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
putrafernanda@DESKTOP-TT1: ~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 21816 1208 ?        Ss   19:32   0:00 /sbin/init
root         2  0.0  0.0  3972 1664 ?        Ss   19:32   0:00 /init
root         7  0.0  0.0  3088 1920 ?        Ss   19:32   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
root        54  0.0  0.2 60704 16704 ?        Ss   19:32   0:00 /usr/lib/systemd/systemd-journald
root       104  0.0  0.0 25372 6272 ?        Ss   19:32   0:00 /usr/lib/systemd/systemd-udev
systemd+  119  0.0  0.1 21456 12544 ?        Ss   19:32   0:00 /usr/lib/systemd/systemd-resolved
systemd+  126  0.0  0.1 91824 7552 ?        Ssl  19:32   0:00 /usr/lib/systemd/systemd-timesyncd
root       180  0.0  0.0  4216 2560 ?        Ss   19:32   0:00 /usr/sbin/cron -f -p
message+  181  0.0  0.0  9628 4688 ?        Ss   19:32   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile
root       191  0.0  0.1 17956 8320 ?        Ss   19:32   0:00 /usr/lib/systemd/systemd-logind
root       193  0.0  0.1 175696 12160 ?       Ssl  19:32   0:00 /usr/libexec/ssl-proc-service -vv
root       197  0.0  0.0  3160 1920 hvsc0 Ss+  19:32   0:00 /sbin/agetty -o -p -- \u --nolc --keep-baud - 115200,38400
systemd+  203  0.0  0.0 222588 5120 ?        Ssl  19:32   0:00 /usr/sbin/rsyslogd -n -lNONE
root       207  0.0  0.1 12020 7088 ?        Ss   19:32   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root       218  0.0  0.0  3116 1792 tty1    Ss+  19:32   0:00 /sbin/agetty -o -p -- \u --nolc - linux
```

```
putraferna@DESKTOP-TTI:~$ ps aux
putraferna 1714  0.0  0.0  9280 4224 pts/0  R+   20:35   0:00 ps aux
putraferna@DESKTOP-TTI:~$ top
top - 20:35:54 up 3:49, 1 user, load average: 0.04, 0.02, 0.
tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,
MiB Mem : 6883.2 total, 6864.1 free, 563.4 used, 326.
MiB Swap: 2848.0 total, 2848.0 free,  0.0 used, 6239.

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM
   1 root         20   0 21816 12488 9336 S  0.0   0.2
   2 root         20   0 3872 1464 1664 S  0.0   0.0
   7 root         20   0 3888 1920 1920 S  0.0   0.0
  54 root        19  -1 66744 16784 15888 S  0.0   0.2
 104 root         20   0 25272  6272  4864 S  0.0   0.1
 119 systemd+    20   0 21456 12544 10368 S  0.0   0.2
 126 systemd+    20   0 91824  7552  6784 S  0.0   0.1
 180 root         20   0 4226 2560 2432 S  0.0   0.0
 181 message+    20   0 9628 4608 4224 S  0.0   0.1
 191 root         20   0 17956 8320 7424 S  0.0   0.1
 192 root         20   0 1756896 12160 10240 S  0.0   0.2
 197 root         20   0 3160 1920 1792 S  0.0   0.0
 203 syslog      20   0 222588 5120 4480 S  0.0   0.1
 207 root         20   0 12820 7888 6784 S  0.0   0.1
 218 root         20   0 3116 1792 1664 S  0.0   0.0
 219 root         20   0 107812 21888 12880 S  0.0   0.3
 317 root         20   0 6664 4880 3712 S  0.0   0.1
 362 putraferna  20   0 20312 11136 9088 S  0.0   0.2
 363 putraferna  20   0 21152 3520 1792 S  0.0   0.1
 388 putraferna  20   0 6072 4864 3328 S  0.0   0.1
 564 polkitd     20   0 388164 7880 6912 S  0.0   0.1
 969 root         20   0 3876 1832 896 S  0.0   0.0
 970 root         20   0 3892 1836 896 S  0.0   0.0
 977 putraferna  20   0 6072 5248 3584 S  0.0   0.1
1471 root         20   0 370884 20352 17536 S  0.0   0.3
1715 putraferna  20   0 9288 5504 3328 R  0.0   0.1

putraferna@DESKTOP-TTI:~$ which bash
/usr/bin/bash
putraferna@DESKTOP-TTI:~$ which gcc
/usr/bin/gcc
putraferna@DESKTOP-TTI:~$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.8.7.2-microsoft-standard-WSL2': No such file or directory
putraferna@DESKTOP-TTI:~$ ls /lib/modules/$(uname -r)/kernel
arch  crypto  fs      lib  net      sound
```

```
putraferna@DESKTOP-TTI:~$ ls /lib/modules/$(uname -r)/kernel
arch  crypto  fs      lib  net      sound
block drivers kernel mm      security virt
putraferna@DESKTOP-TTI:~$ cat /boot/config-$(uname -r) | grep "==" | wc -l # Modules
cat: /boot/config-6.8.7.2-microsoft-standard-WSL2: No such file or directory
0
putraferna@DESKTOP-TTI:~$ lsmod | wc -l
19
putraferna@DESKTOP-TTI:~$ nano procfs_read.c
putraferna@DESKTOP-TTI:~$ gcc -o procfs_read procfs_read.c
procfs_read.c: in function 'main':
procfs_read.c:32:69: warning: implicit declaration of function 'getpid' [-Wimplicit-function-declaration]
   32 | _path, sizeof(proc_path), "/proc/kd/status", getpid());
      |                                     ^~~~~~
putraferna@DESKTOP-TTI:~$ ./procfs_read

=== CPU Information ===
processop      : 0
vendor_id      : AuthenticAMD
cpu family     : 25
model          : 68
model name     : AMD Ryzen 5 6600H with Radeon Graphics
stepping       : 1
microcode      : 0xffffffff
cpu Mhz        : 3293.811
cache size     : 512 KB
physical id    : 0

=== Memory Information ===
MemTotal:      6966480 kB
MemFree:       6359368 kB
MemAvailable:  6398840 kB
Buffers:       2684 kB
Cached:        162620 kB

=== Process Information ===
Name:  procfs_read
Umask: 0022
State: R (running)
Tgid: 1753
Ngid: 0
Pid: 1753
ppid: 977
TracerPid: 0
Uid: 1000 1000 1000 1000
Gid: 1000 1000 1000 1000
putraferna@DESKTOP-TTI:~$
```

9. Building and Booting an Operating System

Praktik 9.1: Analisis Boot Process

Praktik 9.2: Kernel Parameters

Praktik 9.3: Simple Bootloader (Konsep)

```
putrafernanda@DESKTOP-TTU011:~$ dmesg | less
putrafernanda@DESKTOP-TTU011:~$ systemd-analyze
Startup finished in 1.740s (userspace)
graphical.target reached after 1.703s in userspace.
putrafernanda@DESKTOP-TTU011:~$ systemd-analyze blame
369ms landscape-client.service
384ms snapd.seeded.service
327ms dev-sdd.device
266ms snapd.service
214ms my_daemon.service
205ms systemd-udev-trigger.service
185ms user@1000.service
180ms systemd-resolved.service
169ms wsl-pro.service
119ms systemd-timesyncd.service
89ms rsyslog.service
71ms keyboard-setup.service
70ms systemd-journald.service
69ms systemd-tmpfiles-clean.service
67ms polkit.service
64ms systemd-udevd.service
63ms systemd-logind.service
56ms ssh.service
50ms systemd-journal-flush.service
46ms e2scrub_reap.service
45ms user-runtime-dir@1000.service
41ms systemd-tmpfiles-setup.service
36ms dbus.service
35ms dev-hugepages.mount
35ms dev-mqueue.mount
34ms sys-kernel-debug.mount
32ms sys-kernel-tracing.mount
26ms systemd-update-utmp.service
25ms systemd-tmpfiles-setup-dev-early.service
25ms snapd.socket
25ms kmod-static-nodes.service
24ms modprobe@configfs.service
24ms modprobe@dm_mod.service
23ms systemd-binfmt.service
23ms systemd-sysctl.service
```

```
putrafernanda@DESKTOP-TTU011:~$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @1.703s
├─multi-user.target @1.702s
│   └─snapd.seeded.service @841ms +384ms
│       └─basic.target @822ms
│           └─sockets.target @822ms
│               └─snapd.socket @796ms +25ms
│                   └─systemd-resolved.service @606ms +180ms
│                       └─systemd-tmpfiles-setup.service @557ms +41ms
│                           └─systemd-journal-flush.service @501ms +50ms
│                               └─systemd-journald.service @488ms +70ms
│                                   └─systemd-journal.socket @369ms
│                                       └─system.slice @361ms
│                                           └─.slice @361ms

putrafernanda@DESKTOP-TTU011:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
putrafernanda@DESKTOP-TTU011:~$ lsinitramfs /boot/initrd.img-${uname -r} | head -20
Command 'lsinitramfs' not found, but can be installed with:
sudo apt install initramfs-tools-core
putrafernanda@DESKTOP-TTU011:~$ sudo apt install initramfs-tools-core
[sudo] password for putrafernanda:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-glx
  libklib1-mesa libklib1-mesa libklib1-mesa libklib1-mesa libklib1-mesa
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  busybox-initramfs cpio dracut-install initramfs-tools-bin
  klibc-utils klibc zstd
Suggested packages:
  libarchive-dev
```

```
putrafernanda@DESKTOP-TTU011:~$ sudo apt install initramfs-tools-core
Selecting previously unselected package libklibc:amd64.
Preparing to unpack .../4-libklibc-2.0.13-4ubuntu0.1.amd64.deb ...
Unpacking libklibc:amd64 (2.0.13-4ubuntu0.1) ...
Selecting previously unselected package klibc-utils.
Preparing to unpack .../5-klibc-utils-2.0.13-4ubuntu0.1.amd64.deb ...
Unpacking klibc-utils (2.0.13-4ubuntu0.1) ...
Selecting previously unselected package initramfs-tools-core.
Preparing to unpack .../6-initramfs-tools-core-0.142ubuntu25.5_all.deb ...
Unpacking initramfs-tools-core (0.142ubuntu25.5) ...
Selecting previously unselected package zstd.
Preparing to unpack .../7-zstd-1.5.5+dfsg2-2build1.1.amd64.deb ...
Unpacking zstd (1.5.5+dfsg2-2build1.1) ...
Setting up cpio (2.15+dfsg-1ubuntu2) ...
update-alternatives: using /usr/bin/mt-gnu to provide /usr/bin/mt (mt) in auto mode
Setting up libklibc:amd64 (2.0.13-4ubuntu0.1) ...
Setting up busybox-initramfs (1:1.36.1-6ubuntu3.1) ...
Setting up dracut-install (060+5-1ubuntu3.3) ...
Setting up klibc-utils (2.0.13-4ubuntu0.1) ...
No diversion 'diversion of /usr/share/initramfs-tools/hooks/klibc to /usr/share/initramfs-tools/hooks/klibc'i-t by klibc-utils', none
removed.
Setting up zstd (1.5.5+dfsg2-2build1.1) ...
Setting up initramfs-tools-bin (0.142ubuntu25.5) ...
Setting up initramfs-tools-core (0.142ubuntu25.5) ...
Processing triggers for man-db (2.12.0-4build2) ...
putrafernanda@DESKTOP-TTU011:~$ cat /proc/cmdline
initrd=initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_
CRASH_DUMP=1
putrafernanda@DESKTOP-TTU011:~$ sudo nano /etc/default/grub
putrafernanda@DESKTOP-TTU011:~$ bootloader_concept.asm
bootloader_concept.asm: command not found
putrafernanda@DESKTOP-TTU011:~$ nano bootloader_concept.asm
putrafernanda@DESKTOP-TTU011:~$
```


Praktik 10.6: Performance Analysis

[illegible]


```
190.00 0.278258 988 total
buggy.c: command not found
$crackmapexecCMTMP-TF80011-$ nano buggy.c
$crackmapexecCMTMP-TF80011-$ gcc -g nano buggy.c
$crackmapexecCMTMP-TF80011-$ gdb /buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
$crackmapexecCMTMP-TF80011-$ sudo apt install gdb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
libc6-nsvauxv librs-rscent1 libgl1-amdev-dr1
libglapi-mesa libllvm764 libxc6-dr12-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
libbabeltrace1 libc6-dbg libdebuginfod-common
libdebuginfot64 libipt2 libsource-highlight-common
libsource-highlight4t64
Suggested packages:
gdb-doc gdbserver
The following NEW packages will be installed:
gdb libbabeltrace1 libc6-dbg libdebuginfod-common
libdebuginfot64 libipt2 libsource-highlight-common
libsource-highlight4t64
0 upgraded, 8 newly installed, 0 to remove and 2 not upgraded.
Need to get 12.0 MB of archives.
After this operation, 26.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-common all 0.190-1.1ubuntu1 [14.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libbabeltrace1 amd64 1.9.11-2build3 [164 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfot64 amd64 0.190-1.1ubuntu1 [17.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libipt2 amd64 2.6.0-1build1 [45.7 kB]
```

```

putrafernanda@DESKTOP-TTKW011:~$ sudo apt install valgrind
[sudo] password for putrafernanda:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
valgrind is already the newest version (1:3.22.0-ubuntu3).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2c libdrm-radeon1c libgl1-mesa-dri
  libglapi-mesa libllvm764 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
putrafernanda@DESKTOP-TTKW011:~$ nano memleak.c
putrafernanda@DESKTOP-TTKW011:~$ gcc -g -o memleak memleak.c
putrafernanda@DESKTOP-TTKW011:~$ valgrind --leak-check=full ./memleak
==2604== Memcheck, a memory error detector
==2604== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==2604== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==2604== Command: ./memleak
==2604==
==2604==
==2604== HEAP SUMMARY:
==2604==    in use at exit: 400 bytes in 1 blocks
==2604==    total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==2604==
==2604== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==2604==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2604==    by 0x189125: main (memleak.c:4)
==2604==
==2604== LEAK SUMMARY:
==2604==    definitely lost: 400 bytes in 1 blocks
==2604==    indirectly lost: 0 bytes in 0 blocks
==2604==    possibly lost: 0 bytes in 0 blocks
==2604==    still reachable: 0 bytes in 0 blocks
==2604==    suppressed: 0 bytes in 0 blocks
==2604==
==2604== For lists of detected and suppressed errors, rerun with: -s
==2604== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
putrafernanda@DESKTOP-TTKW011:~$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done

```

```
Preparing to unpack .../3-linux-tools-common-6.8.0-85_all.deb ...
Unpacking linux-tools-common (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85.
Preparing to unpack .../4-linux-tools-6.8.0-85_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85 (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85-generic.
Preparing to unpack .../5-linux-tools-6.8.0-85-generic-6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-generic.
Preparing to unpack .../6-linux-tools-generic-6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-generic (6.8.0-85.85) ...
Setting up hwdata (0.379-1) ...
Setting up libnuma1:amd64 (2.0.18-1build1) ...
Setting up libtraceevent1:amd64 (1:1.8.2-1ubuntu2.1) ...
Setting up linux-tools-common (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85 (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Setting up linux-tools-generic (6.8.0-85.85) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
ntrafermand@DESKTOP-TTQU011:~$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
linux-tools-6.6.87.2-microsoft-standard-WSL2
linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
linux-tools-standard-WSL2
linux-cloud-tools-standard-WSL2
ntrafermand@DESKTOP-TTQU011:~$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
linux-tools-6.6.87.2-microsoft-standard-WSL2
linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
linux-tools-standard-WSL2
linux-cloud-tools-standard-WSL2
```